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7 February 1962
Superseding
MIL-T-22555(Wep)
18 August 1960

MILITARY SPECIFICATION

TUBING, STEEL LOW CARBON, ANNEALED, PRESSURE TYPE

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy.

1. SCOPE

1.1 Scope. This specification covers annealed low carbon steel pressure tubing, for use as hydraulic lines.

1.2 Classification. Tubing shall be typed as:

Type I - Seamless.
Type II - Welded and Cold Worked.

2. APPLICABLE DOCUMENTS

2.1 To the extent specified herein, the following specifications and standards, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

P-S-661	Solvent, Dry Cleaning

MILITARY

MIL-C-15205	Corrosion Preventive Compound, Oil, Metal Conditioning

STANDARDS

FEDERAL

Fed. Std. No. 48	Tolerances for Steel and Iron Wrought Products
Federal Test Method Std. No. 151	Metals, Test Methods

MILITARY

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-163	Steel Mill Products, Preparation for Shipment and Storage

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(When requesting specifications, standards and drawings refer to both title and number. Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Supply Depot, 5801 Tabor Avenue, Philadelphia 20, Pennsylvania, Attention: Code CDS.)

3. REQUIREMENTS

3.1 Chemical Composition. The tubing shall conform to the following chemical composition (percentage by weight).

Carbon	0.18 max.
Manganese	0.30 - 0.60
Phosphorus	.040 max.
Sulfur	.050 max.

3.2 Tensile Requirements. The tubing shall conform to the following requirements:

Tensile Strength	55,000 p.s.i. max.
Elongation in 2 inches	35 percent min.
Rockwell Hardness	B65 max.

For tubing with an outside diameter of 3/8 inch or less, or with a wall thickness of .035 or less, or with a combination of both, a minimum elongation of 30 percent is required.

3.3 Dimensions and Tolerances.

3.3.1 Type I tubing. The tolerances given in Table 16ab6 of Federal Standard No. 48 shall apply to outside diameters, inside diameters, and wall thicknesses of Type I tubing.

3.3.2 Type II tubing. The tolerances on outside diameters shall be in accordance with Table 16ab9 of Federal Standard No. 48. The tolerances on wall thicknesses shall be in accordance with Table 16ab6.

3.3.3 Length. Unless otherwise specified in the contract or order, no piece of tubing shall be under the length specified, except that pieces from which test specimens have been cut may be included as part of the order.

3.3.4 Straightness. Unless otherwise specified in the contract, straightness shall be in accordance with Table 16ab12 of Federal Standard No. 48.

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3.4 Flattening. - Tubing shall not develop cracks, tears or other flaws when subjected to the flattening test specified in 4.5.3.

3.4.1 Reverse Flattening. - (For Type II Tubing Only). - The tubing shall show no evidence of cracks or overlaps. When tested in accordance with 4.5.3.1, the tubing shall show no evidence of cracks, lack of penetration, or overlap resulting from flash removal.

3.5 Bending. - Tubing shall not develop any cracks, breaks, or other serious defects when subjected to the bending test specified in 4.5.4.

3.6 Expansion. - Tubing shall expand uniformly and shall show no evidence of cracking when subjected to the expansion test specified in 4.5.5.

3.7 Hydrostatic Pressure. - Each length of tubing shall withstand the internal hydrostatic test pressure specified in 4.5.6 without showing any leaks, permanent dimensional change, or other defect.

3.8 Cold Work. In removing the weld bead (3.10.3), or as a separate operation, the tubing shall be cold worked. This cold working shall result in a minimum reduction in area of 15%, and at least an 8% reduction of wall thickness.

3.9 Heat Treatment. - Annealing shall be accomplished in such a manner as to prevent scaling.

3.10 Workmanship. - All tubing shall be free from injurious defects and shall have a workmanlike finish. Any defect with a depth in excess of 10 percent of the nominal wall thickness or that encroaches on the minimum wall thickness shall be considered injurious. Minor external imperfections such as handling marks, light die or roll marks, shallow pits, and internal imperfections such as mandrel marks, scores, or pits are not considered injurious, providing the imperfections are removable within the tolerances specified for the outside diameter and wall thickness.

3.10.1 Repair of Injurious Defects. - The repair of injurious defects shall be permitted only subject to approval of the purchaser.

3.10.2 Ends. - Unless otherwise specified, tubing shall be furnished with plain ends. The ends shall be smooth and free from burrs.

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3.10.3 Weld Bead. - Type II tubing shall be processed so as to remove the weld bead or any dimensional indication of the presence of the weld.

3.10.4 Latent Defects. - When tubing which has been inspected, tested, and accepted contains subsequently exposed inherent or latent defects covered by this specification, the supplier can be required to replace the defective materi·l without expense to the Government.

3.11 Cleanliness. - The inside of the tubing shall be bright, clean and free from grease, drawing compounds, oxide scale, carbon deposits, and any other contamination that cannot be readily removed with an alkaline cleaner or with solvents conforming to Federal Specification P-S-661.

3.12 Identification Marking. - Each piece of tubing, 1/2 inch or more in diameter and three feet or more in length, shall be legibly marked throughout its length with a row of continuously recurring symbols in the manner set forth below. The symbols shall be applied with a marking fluid which is not soluble in water or petroleum base fluids, or detrimental to the physical properties of the steel. The gap between recurring symbols shall not exceed 3 feet. The marking shall not rub off or be smeared by contact incident to normal handling during shipment and storage.

The following information shall be included:

Code name or trade mark of manufacturer
MIL-T-22555, Type _____
Outside diameter
Wall thickness
Test pressure
Heat number (need appear only once)

The above information should be arranged as in the following example:

Name or Trade Mark MIL-T-22555, Type I 1/2 .049 P7000 Ht 3740

Tubing less than 1/2 inch in diameter or shorter than three feet shall be bundled. Tags giving the above information shall be fastened to each end of each bundle.

4. QUALITY ASSURANCE PROVISIONS

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4.1 Inspection Responsibility. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Inspection Lot. - A lot shall consist of 10,000 feet of tubing or less, all of the same size and wall thickness and made from the same heat of material, from the same annealed batch, or within a four hour period.

4.3.1 Sampling for Chemical Analysis. - One sample for chemical analysis shall be selected from each of two pieces of tubing selected at random from each lot (see 4.2). One sample may be taken from broken tension test specimen. The samples shall consist of not less than two ounces and shall be free from oil and dirt. The samples shall be forwarded to a Government approved laboratory, designated by the Bureau or Agency concerned, for analysis.

4.3.2 Sampling for Tension Test and Hardness Test. - One test specimen shall be selected from each lot (see (4.2) for the tension test and hardness test specified in 4.5.2.

4.3.3 Sampling for Flattening Test. - One piece of tubing shall be selected from each lot (see 4.2) for the flattening test specified in 4.5.3.

4.3.3.1 Sampling for Reverse Flattening Test. One piece of tubing shall be selected from each lot of Type II tubing (see 4.2) for the reverse flattening test specified in 4.5.3.1.

4.3.4 Sampling for Bending Test. - One piece of tubing shall be selected from each lot (see 4.2) for the bending test specified in 4.5.4.

4.3.5 Sampling for Expansion Test. -One piece of tubing shall be selected from each lot (see 4.2) for the expansion test specified in 4.5.5.

4.3.6 Sampling for Hydrostatic Test. - Each piece of tubing shall be subjected to the hydrostatic test specified in 4.5.6.

4.4 Inspection. -

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4.4.1 Dimensional and Visual Inspection. - All tubing shall be subject to dimensional and visual inspection for surface defects and dimensional requirements.

4.5 Test Procedures. -

4.5.1 Chemical Analysis. - The samples selected for test in accordance with 4.3.1 shall be analyzed to determine compliance with 3.1 in accordance with Test No. 111.1 of Federal Test Method Standard No. 151.

4.5.2 Tension Test. - The tension test shall be performed in accordance with Test No. 211.1 of Federal Test Method Standard No. 151. Rockwell hardness test shall be made in accordance with Test No 243.1 of Federal Test Method Standard No. 151.

4.5.3 Flattening Test. - A section of tubing at least three inches in length shall be flattened cold between parallel plates until the distance between plates is three times the wall thickness or twice the wall thickness plus half the internal diameter, whichever is less. When testing welded tubing the specimen shall be so located that the weld is in the area of the sharpest bend radius.

4.5.3.1 Reverse Flattening Test for Type II Tubing. - A section of tubing at least four inches in length shall be cut longitudinally 90 degrees on each side of the weld. The section containing the weld shall be opened and flattened.

4.5.4 Bending Test. - A test section of tubing of sufficient length shall be bent cold 180 degrees around a cylindrical mandrel; the diameter of which is five times the outside diameter of the tubing.

4.5.5 Expansion Test. - A section of tubing of suitable length shall be cut from the tubing selected for the test. It shall be expanded at one end by pressing over a tapered polished steel plug until the tubing is expanded a minimum of 30 percent of outside diameter over a minimum length equal to the outside diameter or one inch of the tubing whichever is greater.

4.5.6 Hydrostatic Test. - Each piece of tubing shall be subjected to an internal hydrostatic test. The hydrostatic test pressure shall be calculated according to Equation (1).

$$P = S \frac{D^2 - d^2}{D^2 + d^2} \qquad (1)$$

Where P = test pressure in psi
S = specified minimum yield strength, psi
D = maximum outside diameter, inches, (nominal OD plus tolerance)
d = maximum inside diameter, inches, (computed as D less twice minimum wall thickness)

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The hydrostatic pressure shall be maintained or exceeded for at least 10 continuous seconds.

4.5.7 Rejection and Retest. - Failure of any test samples to meet the requirements of this specification shall be cause for rejection of the lot represented. The vendor may rework initially rejected lots to remove defectives and resubmit for inspection provided such resubmission has been approved by the purchaser.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. -

5.1.1 Level A.

5.1.1.1 Cleaning. - Unless otherwise specified, the tubing shall be cleaned by any process or combination of processes which will accomplish thorough cleaning without damage to the tubing.

5.1.1.2 Drying. - Immediately after cleaning, tubing shall be thoroughly dried to remove cleaning solutions and/or residual moisture. Drying may be accomplished by any procedure that is not injurious to the tubing.

5.1.1.3 Preservation Application. - The tubing shall be coated inside and outside with a preservative conforming to Type I of Specification MIL-C-15205. All tubing ends shall be capped to exclude foreign matter.

5.1.1.4 Packaging. - Packaging of the tubing shall be in accordance with usual commercial practice.

5.1.2 Level B. - Not applicable.

5.1.3 Level C. -

5.1.3.1 Cleaning, Drying and Preservation. - Cleaning, drying and preservation of the tubing shall be in accordance with 5.1.1.

5.1.3.2 Packaging. - Packaging of the tubing shall be in accordance with the usual commercial practice.

5.2 Packing. -

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5.2.1 Level A. - The tubing shall be packed in accordance with Standard MIL-STD-163.

5.2.2 Level B. - Not applicable.

5.2.3 Level C. - The tubing packaged in accordance with 5.1 shall be packed to afford protection against damage during direct shipment from supply source to the first receiving activity for immediate use. Containers shall comply with the common carrier rules and regulations applicable to the mode of transportation.

5.3 Marking. - In addition to any special marking required by the contract or order, tubing shipments shall be marked in accordance with Standard MIL-STD-163.

6. NOTES

6.1 Intended Use. - Tubing covered by this specification is intended for use as pressure tubing to convey hydraulic oil.

6.2 Ordering Data. - Procurement documents should specify the following.

(1) Title number, and date of specification
(2) Type of tubing
(3) Outside diameter, wall thickness, and length (If possible use sizes shown in Table I.)
(4) Hydrostatic test pressure, if greater than standard psi.

6.3 Weight. - The following data on tubing is included for information only but may serve as a quick check on a shipment to determine its dimensional qualifications. For tubing of 2 inches outside diameter and under, no single length should exceed the calculated weight by more than 10 percent. For tubing above 2 inches outside diameter, no single piece should exceed the calculated weight by more than 18 percent.

In calculating weights use equation (2)

$$W = 10.68\,(D-t)t \quad \ldots\ldots\ldots\ldots\ldots\ldots (2)$$

Where

W = Weight in pounds per foot
D = Outside diameter in inches
t = Wall thickness in inches

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6.4 Levels of Preservation, Packaging, and Packing. - Criteria for use of proper level of preservation, packaging, and packing shall be as follows:

For Level A - This level shall be used for those items which are to be shipped to indeterminate destinations or stored under indeterminate conditions for redistribution anywhere.

For Level C - This level shall be used only when it is definitely known that the packaged item is to be shipped to domestic installations for immediate use at the first receiving activity.

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TABLE I
Standard Sizes of Outside Diameter and Wall Thickness for Pressure Tubing

Outside Diameter (inches)	Wall Thickness (inches)				
1/8	.032	.042	.049		
3/16	.035	.049	.058	.065	
1/4	.035	.049	.058	.065	.083
5/16	.035	.049	.058	.065	.083
3/8	.049	.058	.065	.072	.083
1/2	.042	.058	.083	.095	.109
3/4	.065	.083	.095	.109	.120
7/8	.083	.095	.109	.120	.134
1	.083	.109	.120	.134	.156
1-1/4	.109	.120	.134	.156	.187
1-1/2	.120	.134	.156	.187	.219
1-5/8	.134	.156	.187	.219	.250
1-7/8	.134	.156	.187	.219	.250
2	.134	.156	.187	.219	.250
2-3/8	.156	.187	.219	.250	.313
2-1/2	.156	.187	.219	.250	.313
2-3/4	.187	.219	.250	.313	.375
3	.187	.250	.281	.313	.375
3-1/4	.187	.250	.313	.375	.437
3-1/2	.187	.250	.313	.375	.437

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Notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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